Zhuowei Xu

zhuoweix@andrew.cmu.edu | Website

Education

Carnegie Mellon University Pittsburgh, United States
Master of Science in Mechanical Engineering Sept 2024 – May 2026(Expected)
QPA: 3.9/4.0
Relevant Coursework and Grades: *Mechanics of Manipulation (A), Robot Dynamics and Analysis (A), Machine Learning and Artificial Intelligence for Engineers (A), Numerical Methods in Engineering (A), Modern Control for Robotics (A)*

Shanghai Jiao Tong University Shanghai, China
Bachelor of Mechanical Engineering Aug 2020 – Jul 2024

GPA: 3.5/4.0
Relevant Coursework and Grades: *Calculus I (95.2), Introduction to Engineering (96), Theoretical Mechanics (96), Engineering Material (90), Introduction to Robotics (95), Mechanical Dynamics (91)*

Publications

- Yongzhou Long, Zhuang Zhang, **Zhuowei Xu**, Enlin Gu, Qiujie Lu, Hao Wang, Genliang Chen. "Lightweight and Powerful Vacuum-Driven Gripper with Bioinspired Elastic Spine." [IEEE Robotics and Automation Letters. https://ieeexplore.ieee.org/document/10287554].
- **Zhuowei Xu**, Zilin Si, Kevin Lee Zhang, Moonyoung Lee, Shashwat Singh, Oliver Kroemer, Zeynep Temel. "A Low-Cost Tactile Fingertip Design for Dexterous Robotic Hands." [ICRA 2025 Workshop "Handy Moves: Dexterity in Multi-Fingered Hands". https://openreview.net/forum?id=5kj8uifi2w].
- **Zhuowei Xu**, Zilin Si, Kevin Lee Zhang, Oliver Kroemer, Zeynep Temel. "A Multi-modal Tactile Fingertip Design for Robotic Hands to Enhance Dexterous Manipulation." [Under review. https://arxiv.org/pdf/2510.05382].

Research Experience

Key-Frame Extraction for Long-Horizon Tactile Decision-Making Sept 2025 – Apr 2026
Advisors: Zeynep Temel, Oliver Kroemer, Robotics Institute, Carnegie Mellon University

- Developed a hierarchical tactile decision-making framework that separates manipulation into exploration and execution, using specialized exploratory skills to infer object properties including stiffness, smoothness, weight, and granularity.
- Proposed a top-k tactile key-frame extraction method for compressing long-horizon exploration trajectories into informative observations for downstream policy learning.
- Validated the approach on fine-grained tactile decision tasks, including selecting the smoother side of a sponge via sliding, with generalization to unseen objects.

Tactile-Augmented Imitation Learning Policy for Fine-Grained Manipulation May 2025 – Apr 2026
Advisors: Zeynep Temel, Oliver Kroemer, Robotics Institute, Carnegie Mellon University

- Developed a tactile-augmented diffusion policy framework for dexterous manipulation on a multi-finger robotic hand, integrating vision, force, and vibrotactile sensing for imitation learning.
- Built a multi-modal manipulation system by equipping the robotic hand with custom tactile fingertips, enabling contact-rich policy learning for fine-grained tasks.
- Demonstrated automated grasping of fragile and contact-sensitive objects, where force-aware policies achieved more direct contact perception and more reliable decision-making than vision-only policies in tasks such as pinching potato chips. Showed that tactile augmentation improves first-attempt success rates in dexterous manipulation tasks, including in-hand box opening.

Multi-modal Tactile Fingertip Design for Robotic Hands Oct 2024 – Sept 2025
Advisors: Zeynep Temel, Oliver Kroemer, Robotics Institute, Carnegie Mellon University

- Developed a low-cost, easy-to-make, adaptable, and compact multi-modal fingertip integrating strain gauges for force sensing and a contact microphone for vibrotactile sensing; capable of sensing 2D planar force (0–5 N) and classifying different materials.

- Evaluate across three manipulation tasks with different visual occlusion levels, showing that tactile sensing overcomes occlusion, runs efficiently on lightweight hardware, and captures object properties unreachable to vision based models.

Bionic Vacuum-Driven Gripper with a Elastic Spine Sept 2022 – Nov 2023
Advisors: Genliang Chen, School of Mechanical Engineering, Shanghai Jiao Tong University

- Developed a lightweight and powerful vacuum-driven actuator inspired by snakes' spine and ribs; achieved response time of 320 ms and maximum grasp force > 50 N.
- Designed and 3D-modeled the spine–rib structure and the actuator; devised an integral forming technique by 3D printing PLA ribs on TPU skin, reducing single-chamber manufacture time to < 40 minutes; built a three-fingered gripper and a wireless handheld mobile gripper.

Course Project

Model-Based Trajectory Control and Planning for a Whiteboard Robot Feb 2025 – May 2025
Leading the software development

- Built real-time control and dynamics for a whiteboard-erasing robot; build visual tracking system for target track.
- Implemented a 2D simulation environment and benchmarked PID, MPC, and TVLQR under noise; validated on hardware, with MPC delivering the best tracking.

Design and Simulation of Six-DoF Robot with High Load-to-Weight Ratio Mar 2023 – Jun 2023
Leading the simulation system development

- Designed six-DoF serial robots with a high load-to-weight ratio; implemented control, path planning, and obstacle avoidance using RRT.
- Conducted Simulink–SolidWorks co-simulation; developed a packaged Simscape simulation platform for various environments and loads; implemented an inverted pendulum demo controlled by double-loop PID; selected motors and verified moment requirements.

Professional Experience

Motor Structure Analysis Jun 2022 – Jul 2022
Research Intern | WoLong Electric

- Analyzed load distribution in motor structures and investigated structural optimization strategies for weight reduction.

Underactuated Hand with Dexterous Manipulation Jun 2023 – Sept 2023
Research Intern | Fraunhofer Project Center for Smart Manufacturing at SJTU

- Developed an underactuated finger driven by pouch and face actuators, achieving a lifting capacity of 200 g at 0.1 MPa.
- Improved actuation performance through exoskeleton-based stiffness enhancement and variable-stiffness joint design.

TCP Calibration for a Galvo Mirror–Robot Co-Work System Nov 2023 – May 2024
Research Intern | Shanghai BOCHU Electronic Technology

- Built a galvo mirror–robot co-work system that extends planar galvo processing to 3D surface operations using a 6-DoF robotic arm.
- Developed a kinematic calibration model with error compensation, identifying system parameters from planar images to establish an accurate workspace mapping.

Honors and Awards

- Outstanding Undergraduate Scholarship of SJTU, 2021
- Second Prize, National Physics Competition for College Students in China, 2021
- Second Prize, RoBoMaster "YunHan Cup" Competition of SJTU, 2021

Leadership and Responsibilities

- Graduate Mentor of the CMU SURA (Summer Undergraduate Research Apprenticeship) Project (2025–2026)
- Co-director of the publicity department of the student union (2021–2022)
- Volunteer for freshman registration (Sept 2021, Sept 2022)

- Volunteer teaching during vacations (2020–2023)

Skills

Programming: C++, Python, MATLAB, Julia
Hardware Tools: CAD Design, PCB Design, 3D Printing, Robot Operation System(ROS), Single-chip Development